Chia-Hung Yuan

SENIOR RESEARCH ENGINEER

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About

I am a Senior Research Engineer with a strong track record in on-device machine learning for image and video systems. Experienced in delivering real-time 4K video pipelines (30fps/60fps) on edge platforms. Combines algorithm research, system design, and deployment, with hands-on leadership of cross-functional teams across software and hardware. Published at ICML, AAAI, ISSCC, with multiple granted and pending patents.

Technical Skills

- **Machine Learning**: Deep learning, generative models, video super-resolution, video restoration
- **Computer Vision**: Image Signal Processing (ISP), denoising, ML–ISP integration
- **On-Device Deployment**: Quantization, distillation
- **System Optimization**: Latency, memory footprint, and power efficiency optimization, hardware-aware model design
- **Collaboration**: Cross-functional leadership (SW, MW, driver, HW)

Education

National Tsing Hua University *Hsinchu, Taiwan*
M.SC. IN COMPUTER SCIENCE *Sep. 2019 – Jul. 2021*

- Overall GPA: 4.29/4.30 (top 1%)

National Tsing Hua University *Hsinchu, Taiwan*
B.SC. IN INTERDISCIPLINARY PROGRAM OF ENGINEERING (MATERIAL SCIENCE & QUANTITATIVE FINANCE) *Sep. 2014 – Jun. 2019*

- Overall GPA: 3.95/4.30, Major GPA: 4.01/4.30, CS-related GPA: 4.16/4.30 (top 1%)

Professional Experiences

MediaTek *Hsinchu, Taiwan*
SENIOR RESEARCH ENGINEER *Jun. 2025 – Present*

- Technical lead for the company's first generative video project, leading the design and implementation of a real-time video generation pipeline with strong temporal stability and deployability.
- Contributed to on-device video super-resolution systems supporting 4K@30fps and 4K@60fps, focusing on algorithm development and on-device deployment.
- Collaborated with software, middleware, driver, and hardware teams, contributing to hardware–algorithm co-design.
- Published **2 papers at ISSCC 2025 & 2026** (including **a Highlight Paper**) and filed multiple patents for on-chip image processing systems.

RESEARCH ENGINEER *Jun. 2022 – Jun.2025*

- Researched and deployed deep learning models for image and video enhancement, including denoising, restoration, and super-resolution.
- Designed low-latency, low-power ML architectures for production systems.
- Collaborated with product and system teams on profiling, benchmarking, and optimization.
- Designed and maintained a shared ML codebase to improve development efficiency across projects.

MIT-IBM Watson AI Lab *Massachusetts, USA*
RESEARCH INTERN *Oct. 2021 – Nov. 2021*

- Published **1 paper at AAAI Workshop 2022**.
- Conducted research on meta-learning, neural tangent kernels, and adversarial robustness.

National Tsing Hua University *Hsinchu, Taiwan*
GRADUATE RESEARCH ASSISTANT *Sep. 2019 – Jul. 2021*

- Published **2 papers at ICML 2020 & 2021** on adversarial robustness and generalization.

- Studied neural tangent kernels and robustness, resulting in a **granted U.S. patent**.
- Conducted research in computer vision with emphasis on robust face recognition.

UNDERGRADUATE RESEARCH ASSISTANT *Sep. 2018 – Aug. 2019*

- Designed a model for search engine query-document ranking and achieved **13th place** in MS MARCO passage retrieval.

Advanced Optoelectronic Materials Research Group, National Tsing Hua University *Hsinchu, Taiwan*

UNDERGRADUATE RESEARCH ASSISTANT *Sep. 2017 – Jun. 2018*

- Conducted research on next-generation organic-inorganic hybrid and nano-materials.

University of Tübingen *Tübingen, Germany*

UNDERGRADUATE RESEARCH ASSISTANT *Oct. 2016 – Jul. 2017*

- Conducted research on topography and morphology of solar cell and coupled organic-inorganic nanostructure.

Publications

MADiC: A 3nm 7.4TOPS/mm², 17.4TOPS/W Generative Diffusion Accelerator Enabled by Hardware-Compiler Co-Optimization of Memory Hierarchy and Operator Parallelism *ISSCC 2026*

SHIH-WEI HSIEH, YI-SYUAN CHEN, PING-YUAN TSAI, MING-HUNG LIN, CHIA-YUAN CHENG, LIEN-FENG HSU, PO-HAO HUANG, HUNG-WEI CHIH, PO-HAN CHIANG, CHIA-MING CHANG, MING-HSUAN CHIANG, **CHIA-HUNG YUAN**, SHENG-PO KUO, VISWANATH UGGU, CHUN-KUN CHAN, MING-EN DAVID SHIH, YU-CHENG TSENG, HSIN-PING CHENG, STAN HUANG, CHIA-PING CHEN, SHENKAI CHANG, CHIH-MING WANG, PO-YU YEH, JETT LIU, YUNG-CHANG CHANG, CHI-CHENG JU, YUCHEUN KEVIN JOU

MAE: A 3nm 0.168mm² 576MAC Mini AutoEncoder with Line-based Depth-First Scheduling for Generative AI in Vision on Edge Devices | Paper *ISSCC 2025, **Highlight Paper***

SHIH-WEI HSIEH, **CHIA-HUNG YUAN**, MING-HUNG LIN, PING-YUAN TSAI, YOU-YU NIAN, CHIA-YUAN CHENG, HUNG-WEI CHIH, PO-HAN CHIANG, MING-HSUAN CHIANG, YUAN-JUNG KUO, YU-WEI WE, YI-SYUAN CHEN, PO-HENG CHEN, SANDY HUANG, MING-EN SHIH, CHIA-PING CHEN, ABRAMS CHEN, SHENKAI CHANG, CHIH-MING WANG, PO-YU YEH, JETT LIU, YUNG-CHANG CHANG, CHUNG-YI CHEN, CHI-CHENG JU, CH WANG, KEVEN JOU

Meta Adversarial Perturbations | Paper *AAAI Workshop 2022*

CHIA-HUNG YUAN, PIN-YU CHEN, CHIA-MU YU

Neural Tangent Generalization Attacks | Paper | Video | Code | Competitions *ICML 2021*

CHIA-HUNG YUAN, SHAN-HUNG WU

Adversarial Robustness via Runtime Masking and Cleansing | Paper | Video | Code *ICML 2020*

YI-HSUAN WU, **CHIA-HUNG YUAN**, SHAN-HUNG WU

Patents

Data Poisoning Method and Data Poisoning Apparatus *US Patent 12,105,810*

SHAN-HUNG WU, **CHIA-HUNG YUAN**

Honors & Awards

- **Honorary Member of The Phi Tau Phi Scholastic Honor Society of R.O.C.** (top 3% master's graduands) *2021*
- **Honorary Member of The Phi Tau Phi Scholastic Honor Society of R.O.C.** (top 1% undergraduate graduands) *2018*
- **Academic Achievement Award 3 times** (top 5% students in the class with highest GPA) *2015, 2016, 2018*
- **International Exchange Scholarship** (200,000 NTD/~$7,000) *2016*
- **1st place, Business Case Competition of Seminar on International Trade and Economy** *2016*

Service & Skills

Reviewer	NeurIPS, ICML, ICLR, AAAI, CVPR, IJCAI, CIKM
Teaching Assistant	CS565600 Deep Learning, National Tsing Hua University: Fall 2019, Fall 2020
Languages	Mandarin (Native); English (Fluent); German (Intermediate)